

Data Science

Lecture 07: Data Exploration and Preprocessing

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Converted from LNKB-tex part-en-ds-chp3-data.tex (two source units re-encoded from ISO-8859). Uses Lecture 02's quantiles and covariance throughout.

Module 1

What Kind of Data Is This?

Guiding Question: *The file loads and the numbers compute. Why is that not yet a reason to trust the average?*

Four Attribute Types

Nominal

*categorical,
qualitative*

Labels only: eye
colour, ZIP code, ID.

Meaningful: $=$, $\neq$

Ordinal

*categorical,
qualitative*

Ordered labels:
grades, rankings,
"small/medium/large".

Meaningful: $<$, $>$

Interval

numeric, quantitative

Differences are
meaningful, the zero
point is arbitrary:
temperature in $^{\circ}\text{C}$,
calendar dates.

Meaningful: $+$, $-$

Ratio

numeric, quantitative

A true zero, so ratios
are meaningful:
length, mass, counts,
duration.

Meaningful: $\times$, $\div$

The four types form a hierarchy: each level permits everything the previous one does, plus more. The type is a property of the *measurement*, not of the storage format — which is why a ZIP code stored as an integer is still nominal.

Why the Type Decides the Method

The classic error

Averaging a nominal attribute. The mean of ZIP codes 10115 and 81667 is a number, and it is meaningless.

The subtle one

20 °C is not "twice as warm" as 10 °C — interval scales have no true zero, so ratios do not transfer.

The attribute type determines which **statistics**, which **distance measures** and which **plots** are permissible. Every later chapter depends on it.

Mini-Exercise

Classify each attribute, and name one operation that would be invalid:

1. Customer satisfaction on a five-point scale
2. Year of birth
3. Annual income
4. Preferred payment method

Mini-Exercise — Solution

An attribute's **scale of measurement** licenses the operations you may perform on it. Choosing a statistic before checking the scale is the most common silent error in data analysis.

Mini-Exercise — Solution

1. Ordinal. Averaging is questionable — the gaps between points are not known to be equal.

2. Interval. "Twice the year" is meaningless differences (age) are fine.

3. Ratio. All operations valid zero income is a real zero.

4. Nominal. Only counting and comparison for equality.

An attribute's **scale of measurement** licenses the operations you may perform on it. Choosing a statistic before checking the scale is the most common silent error in data analysis.

Four Shapes of Data

Record data

Objects × attributes
— the familiar table.
Rows are
independent of each
other.

Graph data

Objects carry **links** to
other objects: web
pages, social
networks, citations.

Fields / grids

Objects sit in a
spatial arrangement:
images, sensor
lattices, simulations.

Spatio-temporal

Position **and** time:
monthly land
temperatures, traffic
movement,
trajectories.

The shape decides which methods even apply. A table can be shuffled without loss a grid or a time series cannot — the arrangement *is* information.

Ask what is lost by flattening each non-record type into a table. That question is the bridge to network science and to time-series methods, and it recurs whenever someone loads a CSV.

Three Data Quality Problems

Noise

Random distortion of measured values. Cannot be removed, only modelled or smoothed.

Outliers

Values far from the rest.
Either an **error** to remove or the **signal** you were looking for — fraud, faults, discoveries.

Missing values

Absent entries. Handling depends on **why** they are missing, which the data itself does not tell you.

"Outlier" is not a property of a data point; it is a judgement about a model of what is normal. Deleting outliers deletes evidence against that model.

Mini-Exercise

A sensor dataset has 8% missing temperature readings.

1. Give two mechanisms that could produce them.
2. For each, is deleting those rows acceptable?
3. What would you need to know to decide?

Mini-Exercise — Solution

The shape of the data decides which methods apply the **reason** for a quality problem decides how it may be handled. Neither is visible in the file.

This is Principle 3 in its most practical form: a preprocessing decision is a modelling assumption, and here an untestable one. Students should leave able to ask "why is this missing?" before "how do I impute it?".

Mini-Exercise — Solution

1. Random transmission dropouts — or the sensor failing **when it is very cold**.

2. Dropouts: deletion is roughly harmless. Cold-failure: deletion silently removes the coldest days and biases every statistic afterwards.

3. The **mechanism** — whether missingness depends on the unobserved value itself. The data alone cannot answer this domain knowledge must.

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Module 2

Seeing the Data

Guiding Question: *Summary statistics answer the questions you thought to ask. What shows you the ones you did not?*

Visualization Is Everywhere

Newspaper tables, subway maps, weather charts, stock graphs, a 3-D reconstruction of an injured knee, demographic bar charts, social-network layouts.

What do these have in common? All of them present data in a form the eye processes faster than the equivalent numbers.

Question-first (P1). Let students name the common property before it is stated; "you can see it at a glance" is the answer, and the next slide gives the reason.

Why It Works

Bandwidth

Vision is the human sense with the highest data rate — pattern detection runs in parallel and pre-attentively.

Interaction

A picture is a natural surface for exploring: zoom, filter, select, compare.

Discovery

Visualization surfaces the **unexpected** — the structure nobody thought to compute a statistic for.

Summary statistics answer questions you already asked. A plot lets the data ask you one — which is exactly what Anscombe's quartet demonstrated.

Visualization is not decoration for results; it is an **instrument of analysis**, and the only one that can show you what you did not think to measure.

Deliberate callback to L01 and L02 (spiral): the same argument, now with the tools to act on it. This unit is the hinge between "data does not speak for itself" and the concrete plot types that follow.



The Boxplot

A visual summary of a distribution built from quartiles:

The box

Spans Q_1 to Q_3 — the interquartile range, the middle half of the data.

The line

The **median** inside the box. Its position shows skew.

The whiskers

Minimum and maximum, or (Tukey) the most extreme points within $1.5 \times \text{IQR}$. Points beyond them are drawn as **outliers**.

The $1.5 \times \text{IQR}$ rule is a **convention**, not a test. It flags candidates for inspection, not errors for deletion.

Speaker notes

Boxplots are built entirely from the quantiles the students defined in Lecture 02 — say so. Their strength is comparison: ten boxplots side by side are readable where ten histograms are not.



The Histogram

What it shows

The **shape** of a distribution: modality, skew, gaps and clusters that a boxplot hides entirely.

What it hides

Everything depends on the **bin width**. Too wide and structure vanishes too narrow and noise looks like structure.

A bimodal distribution and a unimodal one can produce the same boxplot. Always look at one histogram before trusting a five-number summary.

Mini-Exercise

Two datasets have identical minimum, Q_1 , median, Q_3 and maximum.

1. Must they have the same histogram?
2. Sketch two distributions with the same boxplot but a visibly different shape.
3. What does this repeat from an earlier lecture?

Mini-Exercise — Solution

A boxplot compares distributions compactly; a histogram shows their shape. **Neither alone is sufficient**, and the boxplot's outlier rule is a convention you may overrule.

Mini-Exercise — Solution

1. No. Quartiles fix five points the mass between them can be arranged freely.

2. A uniform spread versus two tight clusters at the ends of the box — same quartiles, opposite stories.

3. Anscombe's quartet: summaries agree, the data does not. The lesson is scale-free.

A boxplot compares distributions compactly; a histogram shows their shape. **Neither alone is sufficient**, and the boxplot's outlier rule is a convention you may overrule.

The Scatterplot

An X–Y plot of two dimensions against each other — the standard instrument for inspecting a **relationship** rather than a distribution.

With more than two dimensions there are four options:

Subsetting

Let the user choose which dimensions to display.

Reduction

Project to fewer dimensions (see preprocessing, and projections later).

Embedding

Map extra dimensions to other channels: colour, size, shape.

Multiple displays

Superimpose, or juxtapose in a **scatterplot matrix**.

The four options are exhaustive and worth memorising — every high-dimensional visualization technique is one of them or a combination. Colour and size are channels with far lower precision than position use them for secondary variables.

What a Scatterplot Reveals

Linear association, non-linear structure, clusters, outliers and heteroscedasticity — most of which no single correlation coefficient reports.

A correlation of 0 is consistent with a perfect parabola. The scatterplot shows it the coefficient does not.

Tufte's Principle

Maximise the data-ink ratio: every drop of ink on the page should carry information. Delete anything that does not — heavy gridlines, boxes, redundant tick marks, decorative fills.

Range frames apply the principle to the axes: draw each axis only over the interval the data actually spans, so the frame itself reports the minimum and maximum.

The exercise: take a finished chart and ask of each element, *what would I lose by deleting this?*

A scatterplot shows the structure a coefficient summarises away. Beyond two dimensions you must **subset, reduce, embed or juxtapose** — and every mark you draw should earn its ink.

Tufte's rule is a good design heuristic and an over-applied dogma note that some redundancy aids reading under time pressure. The point students should keep is intentionality: each element defended, not each element removed.

When a Line Is Justified

Connecting points with a line asserts an **order** between them — and often an interpolation.

Data requirement: one or more quantitative value attributes on a comparable scale, plus one **ordered** key attribute (usually time).

Connecting nominal categories with a line claims an order that does not exist. The line is a statement, not a decoration.

This is the attribute-types unit paying off immediately: the permissible plot follows from the scale of the key attribute. Bar charts for nominal, lines for ordered.

Aspect Ratio Is Not Cosmetic

Humans judge **angles** far more accurately than absolute slopes, so the shape of the plotting area changes what the reader concludes. The same series looks flat in a wide frame and dramatic in a tall one.

Banking to 45° chooses the aspect ratio so that the average line segment sits near 45 degrees — the angle at which slope differences are most discriminable.

Additional channels

Colour and line type distinguish series — but only for a handful of them.

Multiple Series

Overlaid

Best for comparing a few series directly against each other.

Stacked

Each series is drawn on top of the previous one. Shows the total well — and makes individual series **hard to read**, since each sits on a moving baseline.

Sorted / small multiples

One panel per series, ordered by some property. Scales to many series where overlaying fails.

A line asserts order and interpolation the **aspect ratio** decides what the reader sees and stacking trades the readability of the parts for the readability of the whole.

The stacked-plot critique is worth demonstrating: ask students to read the third band's trend off a stacked chart — they cannot, because the baseline moves. Small multiples are the usual better answer.

Module 3

Preprocessing

Guiding Question: *Most of a project is spent before the model. What is actually being decided in that time?*

Aggregation

Combine several objects or attributes into one — customer purchases summarised by store, by month, by price band.

Gains

Fewer objects to process **less variation**, hence more stable estimates the natural operation of a data warehouse.

Costs

Detail is gone, irreversibly. Effects that exist only within a group become invisible.

Speaker notes

Aggregation is where Simpson's paradox lives: a trend can reverse when groups are pooled. Worth one sentence — students will meet it in dashboards long before they meet it in a statistics course.



Sampling

Work with a subset of the objects: random sampling with or without replacement, or **stratified** sampling that preserves group proportions.

Sampling always loses information. The question is whether it loses anything that matters — a sufficiently large sample retains the overall structure.

The sine-wave example:
sample too sparsely and
the reconstructed signal is
not merely rougher, it is
wrong.

Dimensionality Reduction

The curse of dimensionality. As the number of attributes grows, the data becomes sparse: distances between points concentrate, neighbourhoods lose meaning, and the volume to be covered grows exponentially.

The problem

Many features are irrelevant, noisy or correlated.

The cost

More computation, more parameters, more variance — and weaker distance-based methods.

The response

Feature selection keeps a subset **feature extraction** builds new combined dimensions.

Every preprocessing step **trades information for tractability**. The trade is usually worth it — but it is a modelling decision, and it should be recorded as one.

Point forward: projections (PCA, MDS) are the formal treatment of feature extraction, and clustering suffers directly from the curse. This unit is the motivation both later lectures rely on.

